

The Agentic Loop.

A practitioner's guide to designing, governing, and scaling closed-loop AI agent systems in production — from single-agent workflows to orchestrated enterprise fleets.

40%

of enterprise software is projected to embed task-specific agents by end of 2026, up from under 5% in 2025

**79% /
11%**

of organizations report some agent adoption, but only 11% have reached production

171%

average 18-month ROI reported by enterprises deploying agentic AI with proper governance

5

composable workflow patterns underpin nearly every production agent architecture

For:

Enterprise architects, platform engineering & AI governance leads

Edition

June 2026

GUIDE MAP

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SECTION 01

Why Loops Matter

The shift from prompt/response to goal/loop/outcome is the single most consequential architectural change in enterprise AI since the chatbot.

Most organizations' first encounter with generative AI is a single-turn interaction: a person writes a prompt, a model returns a response, and a human decides what happens next. That pattern is simple to reason about and easy to govern — but it caps the system's usefulness at the speed of human attention. An agent that must be re-prompted after every step cannot debug its own code, reconcile a multi-step finance close, or work through a backlog while the team sleeps.

An **agentic loop** breaks that ceiling. Instead of a person driving every step, the agent is given a goal, a set of tools, and a way to check its own work, and it iterates — plan, act, observe, adapt — until the objective is met, a budget is exhausted, or it hits a condition that requires a human. The loop is the mechanism that turns a language model from a chatbot into something that can actually do work.

Prompt → Response

The human drives every step

- Human is the bottleneck on every decision
- Constant context-switching between tasks
- Doesn't scale past one operator's attention
- Work stops the moment the human stops

Goal → Loop → Outcome

The agent drives execution

- Plans the path, executes, checks, retries
- Continuous progress without re-prompting
- Consistent quality via repeatable verification
- Scales to workloads no single human could track

The state of adoption: Gartner projects that 40% of enterprise software applications will integrate task-specific AI agents by the end of 2026, up from less than 5% in 2025. Yet a March 2026 enterprise architecture study found that while 79% of organizations report some AI agent adoption, only 11% are in production and just 2% have deployed at full scale — a gap that almost always traces back to architecture and governance, not model capability.

Continuous progress

Work advances on a schedule, not on a human's availability.

Consistent quality

A repeatable verification step replaces ad hoc human review.

Scalable workflows

One architecture generalizes across many tasks and teams.

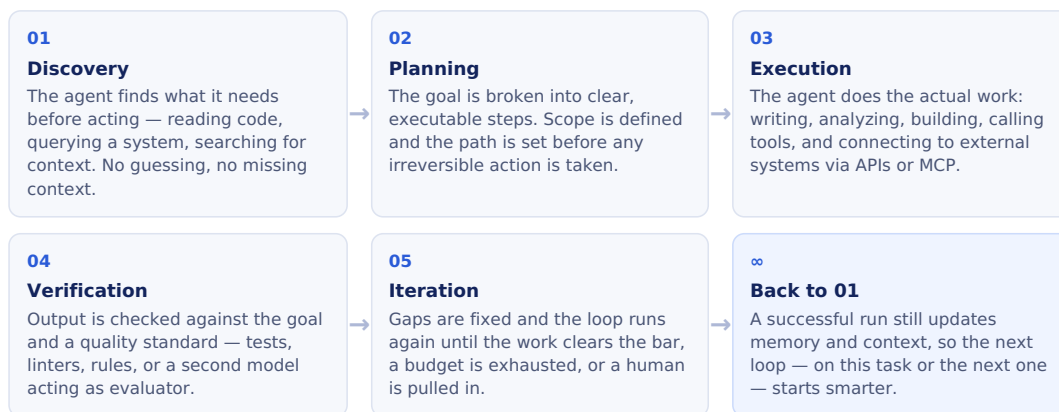
Enterprises deploying agentic AI with proper governance report average ROI of 171% within 18 months. The phrase to hold onto is *with proper governance*: the rest of this guide treats the loop and its guardrails as a single design problem, not two separate workstreams.

SECTION 02

The Canonical Agent Loop

Every production agent — regardless of framework, model provider, or use case — converges on the same five-phase cycle.

Strip away the framework branding and nearly every working agent runs the same loop: it figures out what it's missing, breaks the goal into steps, does the work, checks the work against a standard, and fixes what failed before trying again. This is the architecture Oracle's developer team and Anthropic's own internal tooling both converge on, and it is the right mental model to design against before picking any framework.



This pattern holds at every scale: a coding agent fixing a failing test, a research agent gathering and synthesizing sources, and a finance agent reconciling a ledger are all running the same five phases, just with different tools and different definitions of "done." A useful way to confirm an architecture is sound: the agent reasons about what it needs, calls a tool, gets a result, decides whether to continue or stop — and across every model provider, tool integration follows that same shape. Tools are defined with a name, a description, and a parameter schema; the model decides whether to call one; the system executes it and returns a result; the model decides whether to loop again or return a final answer.

Design implication: Treat each phase as independently testable. A loop that fails in production almost always fails because one phase was collapsed into another — planning skipped in favor of immediate execution, or verification reduced to "the agent said it was done." Each phase should be inspectable on its own.

SECTION 03

Open Loops vs. Closed Loops

The single architectural decision with the largest effect on cost, predictability, and blast radius.

An **open loop** gives the agent a goal and lets it explore with few constraints on retries, scope, or termination. An **closed loop** gives the agent the same goal but bounds it with explicit budgets, validation checkpoints, and a defined goal state. The difference looks subtle in a design doc and is enormous in a production incident.

Open Loop — the agent wanders

- Good for exploratory, ambiguous tasks
- Unlimited or loosely bounded retries
- High and unpredictable token usage
- Can drift away from the original objective

Open-loop agents have been observed burning upward of 2M tokens in a single run when no budget or stopping condition was set.

Closed Loop — the agent stays on track

- Goal and success criteria defined up front
- Validation checkpoints between phases
- Hard budget limits on tokens, calls, and time
- Predictable, bounded outcomes

Closed loops are what fix the cost and drift problem — this is the pattern almost all production agents now use.

In practice, "closed" does not mean rigid. The best production systems are closed at the boundary — hard limits on spend, scope, and irreversible actions — while remaining open within those boundaries, so the agent still has room to explore alternate approaches when the first one fails. The discipline is in defining the boundary explicitly rather than discovering it after an incident.

CONTROL	WHAT IT BOUNDS	TYPICAL IMPLEMENTATION
Token / cost budget	Total spend per run	Hard ceiling enforced by the orchestration layer, not the model
Step / iteration cap	How many loop cycles run	Counter checked before each new planning phase
Scope allowlist	Which tools, files, or systems can be touched	Tool registry scoped per task, enforced at the gateway
Validation checkpoint	Whether to proceed to the next phase	Tests, linters, schema checks, or an evaluator model
Termination condition	When the loop ends	Goal met, budget exhausted, or escalation triggered

SECTION 04

The Five Composable Workflow Patterns

Anthropic's reference taxonomy, distilled from work with production customers, remains the clearest map of agentic architecture available. Start here before reaching for a framework.

Anthropic draws a deliberate distinction between two categories of agentic system. **Workflows** are systems where the LLM and tools are orchestrated through predefined code paths — the control flow is fixed, even if the content at each step is generated. **Agents**, by contrast, are systems where the LLM dynamically directs its own process and tool use, retaining control over how a task gets done. Across both categories, the most successful production implementations are not the ones using the most sophisticated frameworks — they are the ones built from simple, composable patterns.

Five patterns cover nearly every production agentic system in use today: **prompt chaining**, **routing**, **parallelization**, **orchestrator-workers**, and the **evaluator-optimizer loop**. They are composable — production systems frequently combine two or three of them inside a single workflow.

1 · Prompt Chaining

2 · Routing

3 · Parallelization

4 · Orchestrator-Workers

5 · Evaluator-Optimizer

PATTERN 1

Prompt Chaining

Decomposes a task into a fixed sequence of steps, where each LLM call processes the output of the one before it. Trades latency for accuracy by making every individual call an easier task.

Use when: the task decomposes cleanly into fixed subtasks — e.g. drafting marketing copy, then translating it; or writing an outline, validating it against criteria, then writing the full document from the validated outline.

PATTERN 2

Routing

Classifies an input and directs it to a specialized downstream task, separating concerns so each path can be optimized independently rather than forcing one prompt to handle every case well.

Use when: inputs fall into distinct categories that are better handled separately and can be classified reliably — e.g. routing support tickets to billing vs. technical paths, or routing easy queries to a smaller, cheaper model and hard ones to a more capable one.

PATTERN 3

Parallelization

Runs independent LLM calls simultaneously. *Sectioning* splits a task into independent subtasks run in parallel; *voting* runs the same task multiple times to combine outputs and raise confidence.

Use when: speed matters and subtasks are genuinely independent, or when diverse perspectives on the same input improve reliability — e.g. parallel content moderation checks, or multiple independent code reviews of the same diff.

PATTERN 4

Orchestrator-Workers

A central LLM dynamically breaks a task down and delegates subtasks to worker LLMs, then synthesizes their results. Unlike parallelization, subtasks aren't pre-defined — the orchestrator determines them from the specific input.

Use when: complexity is unpredictable — e.g. multi-file coding changes where the number and nature of affected files can't be known in advance, or research tasks that require gathering and synthesizing information from multiple sources.

PATTERN 5

Evaluator-Optimizer Loop

One LLM call generates a response while a second LLM call — the evaluator — provides structured feedback in a loop, refining the output until it clears a quality bar. This is the pattern that most directly implements "the loop" inside a single task: an optimizer proposes, an evaluator critiques, and the cycle repeats until the evaluator is satisfied or a retry limit is hit.

Use when: there is a clear evaluation criterion and iterative refinement provably improves output — e.g. literary translation where a second pass catches nuance the first missed, or complex search tasks requiring multiple rounds of searching and analysis to gather complete information.

Choosing a pattern

PATTERN	BEST FOR	PRODUCTION EXAMPLE
Prompt Chaining	Fixed, cleanly decomposable steps	Generate copy → translate → format for channel
Routing	Distinct, reliably classifiable input types	Support query triage by category and difficulty
Parallelization	Independent subtasks; latency-sensitive	Multi-perspective content moderation, parallel reviews
Orchestrator-Workers	Unpredictable, input-dependent complexity	Multi-file coding agents; open-ended research
Evaluator-Optimizer	Clear quality bar, iterative refinement pays off	Code review with revision; document quality loops

On frameworks: the most consistent finding across production deployments is that teams overestimate how much abstraction they need. Frameworks can help you start quickly, but they often obscure the underlying prompts and responses, making them harder to debug, and can tempt teams toward complexity a simpler setup would handle just as well. Before reaching for an orchestration framework, implement the relevant pattern directly against the model API. Add abstraction only when it removes real complexity you've already hit.

Three principles for implementation

Maintain simplicity

Add agentic complexity only when a simpler workflow demonstrably falls short.

Prioritize transparency

Explicitly surface the agent's planning steps rather than hiding reasoning inside a black box.

Craft the ACI

Invest in the agent-computer interface — tool docs and testing — as carefully as a human-facing UI. See Section 08.

SECTION 05

Single Agent vs. Agent Fleet

A single agent attempting an entire enterprise workflow in one reasoning loop is brittle by construction. Knowing when to fan out is a core architecture decision.

A single agent in one context window accumulates bloated context, mixed responsibilities, and a single point of failure as task complexity grows. The fix is not a bigger model — it's multi-agent orchestration: coordinating a small swarm of specialized agents toward a shared goal, each with a narrow context and a clear responsibility.

Single Agent

One agent, one context, one decision maker

- Coding tasks scoped to one repo or module
- Research with a narrow, well-defined question
- Content generation against a fixed brief

FOCUS: SIMPLICITY

Agent Fleet

Specialized agents, shared objective, orchestrated workflow

- Large or unfamiliar codebases spanning many files
- Deep, multi-source research and synthesis
- Enterprise systems with many integrated steps

FOCUS: SCALE

Anthropic's own multi-agent research system is a useful reference design: a **lead agent** analyzes the incoming query, develops a search strategy, and spawns specialized subagents that operate in parallel, each acting as an intelligent filter that iteratively uses tools to gather information before reporting back a condensed result to the lead agent, which then synthesizes the final answer. The lead agent never sees the subagents' raw intermediate work — only their distilled findings — which is what keeps its own context manageable as the fleet scales.

Architectural rule of thumb: orchestrator-worker (Pattern 4 in Section 04) is the load-bearing structure of nearly every effective multi-agent fleet. A central orchestrator decomposes the task, dispatches narrowly-scoped workers, and reconciles their output — it does not let workers talk to each other directly, which is what keeps the system debuggable as it grows.

SIGNAL	LEAN TOWARD SINGLE AGENT	LEAN TOWARD FLEET
Scope	Bounded, well-understood	Large, spans multiple systems or files
Predictability	Subtasks are knowable in advance	Subtasks depend on what's discovered
Latency tolerance	Needs a fast, single-pass answer	Can tolerate parallel exploration time
Failure cost	Low — easy to retry the whole task	High — isolating failures to one worker matters

SECTION 06

The Six Building Blocks of Production Loops

Patterns describe control flow. These six components are what actually turn a pattern into something that runs unattended, safely, in an enterprise environment.

Automations

The loop runs on a schedule or trigger, not on a person remembering to kick it off. This is what separates a demo from a production system.

Worktrees

Parallel agents operate in isolated working copies so concurrent runs don't collide on the same files or state.

Skills

Project and domain knowledge is written once, as a reusable artifact, and read by the agent on every loop instead of being re-explained in every prompt.

Plugins & Connectors

Standardized connections out to PRs, tickets, chat, and external systems — increasingly via the Model Context Protocol (MCP), now the de facto open standard for agent-to-tool integration.

Subagents

The agent that does the work and the agent that checks the work are never the same instance — a maker/checker split that prevents an agent from grading its own homework.

Memory

Durable state lives outside the conversation window, on disk or in a store, so the loop never forgets what it learned in a prior run.

Why Skills matter for consistency: teams that rewrite the same long context block into every prompt see behavior drift over time as the prompt is copy-edited inconsistently across users. Defining a skill once, as a versioned artifact the agent reads on every loop, is what keeps behavior consistent at scale — the same logic that makes a documented runbook more reliable than tribal knowledge.

Of the six, **memory** and **subagents** are the two most often skipped in early prototypes and the two most responsible for production incidents when missing. A loop with no external memory re-derives context from scratch every run, burning tokens and re-making mistakes it already fixed once. A loop where the same model checks its own output has no real verification step — it has a second opinion from someone with every incentive to agree with the first.

SECTION 07

The Quality Gate

No gate is a slop machine. Build the gate from things the agent can't argue with.

Every closed loop needs exactly one moment where output is checked against a standard before it ships. That moment is the quality gate, and its defining property is that it must be a check the agent cannot talk its way past — a deterministic test, not a request for the agent's own opinion of its work.



On failure: a failed gate routes back into the loop, not to a human by default. The agent receives the specific failure (which test, which line, which rule) and re-enters execution. Escalate to a human only after a bounded number of failed attempts, or when the failure pattern itself signals something the gate wasn't designed to catch.

GATE TYPE	CATCHES	WHERE IT RUNS
Tests	Functional regressions, broken logic	Unit / integration test suite
Linters	Style violations, common bug patterns	Static analysis, pre-commit
Type checks	Interface mismatches, null-safety issues	Compiler / type checker
CI / CD	Build failures, integration breakage	Pipeline, before merge or deploy
Security checks	Vulnerable dependencies, secrets, injection risk	SAST/DAST scanners, dependency audit

The evaluator-optimizer pattern from Section 04 is the natural home for a quality gate when the standard is more nuanced than a pass/fail test — for example, judging the tone of a customer-facing email or the completeness of a research summary. In those cases the evaluator is a separate model call (or separate agent instance) with explicit rubric criteria, never the same call that produced the work.

SECTION 08

The Agent-Computer Interface

Tool design is the highest-leverage, most under-invested part of most agent architectures.

Just as human-computer interaction designers spend significant effort designing good UIs, building good agent-computer interfaces (ACI) deserves equal care. Tools are how an agent perceives and acts on the world; a poorly documented or ambiguous tool produces unreliable agent behavior no amount of prompt engineering can fully compensate for.

What a well-designed tool looks like

- A clear name and a description written as if for a new engineer on the team
- Parameters with explicit types, constraints, and example values
- Error messages that tell the agent what to try differently, not just that something failed
- Outputs in a format the model parses reliably — structured, not ad hoc prose

Common ACI failure modes

- Two tools with overlapping purposes the model can't reliably distinguish
- Free-text parameters where a constrained enum would remove ambiguity
- Silent failures that return an empty result instead of a clear error
- Tools that require knowledge the model has no way to acquire

Connectivity itself has converged on a standard: the **Model Context Protocol (MCP)**, developed by Anthropic, has become the de facto open standard for agent-to-tool integration in 2026, giving agents a consistent way to discover and call tools across ERPs, CRMs, ticketing systems, and internal services without a bespoke integration for every pairing of agent and system.

Test tools the way you test code. Before trusting a tool in a production loop, run the agent against it with a held-out set of realistic tasks and read the transcripts. Most ACI problems surface immediately once you watch an agent try, fail, and guess at why — and are invisible if you only read the tool's schema.

SECTION 09

Governance & Autonomy Tiers

The choice is not between full human control and full autonomy. It's calibrating oversight to the reversibility and consequence of each action.

Treating human oversight as binary is the most common governance mistake. Organizations that require human-in-the-loop approval for every agent action defeat the value of agentic AI; organizations that apply no oversight to high-risk agents expose themselves to regulatory liability and operational harm. The right model calibrates oversight to what's actually at stake in a given action.

Human-in-the-loop

A human must approve a significant agent action *before* it executes. Appropriate for irreversible, high-consequence, or low-confidence actions — e.g. an outbound payment, a public-facing communication, or a change to production infrastructure.

Human-on-the-loop

The agent acts autonomously while a human monitors behavior and reviews logs retrospectively. Appropriate for reversible, low-consequence, well-tested actions — e.g. ticket triage, internal data lookups, draft generation.

Autonomy levels by system integration

INTEGRATION	READ ACCESS	WRITE / CONSEQUENTIAL ACTIONS
ERP (SAP, Oracle)	Master data via least-privilege API access	Purchase orders, record updates require human approval at minimum
CRM (Salesforce, HubSpot)	Account history, deal stage	Customer writes require confidence thresholds and full audit logging
ITSM (ServiceNow, Jira)	Ticket triage and classification	Ticket creation/updates can run autonomously; on-call escalation has explicit triggers
HR (Workday, SuccessFactors)	Onboarding workflows	Account creation and training assignment, with defined escalation paths

Most enterprise use cases in 2026 still operate at the lower end of the autonomy spectrum for high-risk actions — full autonomy is reserved for narrow, well-validated, reversible workflows. Calibrate ambition to the maturity of your quality gate (Section 07), not the other way around.

Build compliance as an architectural feature from day one, not as something bolted onto an existing deployment after an incident or an audit. That means audit trails, decision logs, and role-based access embedded in the orchestration layer itself, plus explicit, written policy on which action classes require approval, how long decision data is retained, and who holds override authority.

SECTION 10

The Regulatory Landscape

Regulation written for predictive models is being stretched to cover systems that act. Architects should track the gap, not just the letter of each law.

Most existing AI governance frameworks were built to assess risk at training or deployment time. Agentic risk manifests in *execution*, not configuration — the relevant question is not what the model can do, but what the agent actually does across a full, multi-step run. Several major frameworks bear directly on enterprise agent deployments in 2026.

FRAMEWORK	WHAT IT REQUIRES	STATUS / DEADLINE
EU AI Act	Effective human oversight for high-risk systems; traceability, documentation, and the ability to stop, correct, or override any autonomous action	High-risk obligations phasing in through 2027; broader provisions already enforceable
NIST AI RMF + agent profile	The Govern, Map, Measure, Manage functions, extended with an agent-specific profile covering identity, authentication, and containment boundaries	Agent-focused profile from NIST's CAISI initiative planned for late 2026
Colorado AI Act	Oversight obligations for AI systems making consequential decisions about Colorado residents	Deadline June 30, 2026
ISO/IEC 42001	AI management system standard; increasingly used to govern third-party agent vendors and supply-chain risk	Voluntary certification, growing enterprise adoption

The liability question remains genuinely unsettled. When an autonomous agent takes a harmful action — an unauthorized trade, an erroneous communication, an infrastructure change — responsibility may fall on the model developer, the deploying organization, or the end user depending on jurisdiction and contract terms. As a working assumption, treat the deploying organization as the default bearer of liability and architect logging and approval controls accordingly; confirm the specific allocation with legal counsel rather than relying on general guidance here.

Minimum architectural requirements emerging across frameworks

Visible, not black-box Decision logic must be documented and inspectable — regulators describe this as replacing the black box with a "glass box."	Stoppable, always Every agent needs a mechanism to halt, correct, or override its operation at any point in a run.
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This guide is not legal advice. Regulatory deadlines and obligations shift quickly — verify current requirements against primary sources and counsel before finalizing a compliance posture.

SECTION 11

Security & Identity

Agents are not users and not service accounts. They need their own identity model, with its own threats and its own controls.

The deployment of autonomous agents introduces failure modes that don't map cleanly onto traditional application security: data leakage through extensive context windows, indirect prompt injection from content the agent reads rather than from its operator, and identity spoofing that exploits the gap between who an agent claims to be acting for and what it's actually been authorized to do.

Primary risk patterns

- **Indirect prompt injection** — malicious instructions embedded in a document, email, or web page the agent reads as data
- **Tool misuse via deceptive framing** — an agent is talked into using a legitimate tool for an illegitimate purpose
- **Identity spoofing** — actions taken under a false or over-broad identity
- **RBAC drift** — gaps between intended access policy and what the agent can actually reach

Core controls

- **Distinct agent identity** — never reuse a human's credentials or a shared service account
- **Least privilege** — scope tool and data access to exactly what the task requires
- **Gateway-level enforcement** — policy decisions made outside the model, not requested of it
- **Adversarial testing** — red-team for injection and tool misuse before and after deployment

Enforce at the gateway, not in the prompt. The control plane — not the system prompt — should be what rejects an out-of-scope action. If an agent attempts a call outside its authorized scope, the gateway should reject the API call and log the policy violation, independent of anything the model itself decided. A determined or confused agent should never be the last line of defense for its own permissions.

Govern agents within your existing AI risk program rather than standing up a parallel structure. The four functions of the NIST AI Risk Management Framework — Govern, Map, Measure, Manage — apply directly to agentic systems; what changes is the granularity: inventory every agent identity, every tool grant, and every standing credential the way you would any other privileged service.

SECTION 12

Observability & the Audit Plane

Production environments require more than clever code — they require reliability, observability, and a habit of continuous improvement.

An **audit plane** is the system that logs every tool invocation, every routing decision, and the provenance of every final output, so an architect can forensically reconstruct exactly how an agent reached a given result. Without it, a failed run is a mystery; with it, a failed run is a debugging exercise like any other.

What to log on every run

CATEGORY	CAPTURED DETAIL
Prompts & outputs	Full input/output pairs at every step, with template or skill version
Tool invocations	Tool name, arguments, return value, latency, and success/failure
Routing & planning decisions	Which path the agent chose and the reasoning that produced it
Human feedback	Approvals, rejections, and explicit ratings tied to the specific run
Timestamps & identity	Who or what triggered the run, under which credentials, and when

If an agent fails, tracing the decision chain backward is usually how the root cause is found — which is only possible if every step was actually recorded, not summarized after the fact by the agent itself. For regulated industries, this level of structured logging is not a nice-to-have; it is what auditors will ask for directly.

Evaluation loops
Automated regression tests built specifically for agentic logic, scoring new runs against a baseline rubric so quality regressions are caught the same way a code regression would be.

Hard stopping conditions
Caps on continuous inference cycles, independent of the agent's own judgment, to prevent runaway cost or a stuck loop from running indefinitely.

A large share of practitioners now use human-in-the-loop evaluation specifically to validate agent performance over time — not just at launch. Treat the first production deployment as the start of an ongoing evaluation practice, not the finish line.

SECTION 13

The Loop Engineering Maturity Model

Maturity is not about which model an organization uses — it's how much of the loop has been engineered versus how much still depends on a person driving each step.

LEVEL 1

Prompter
Reactive. One task at a time. A person writes a prompt, reads the response, and decides the next prompt by hand. No loop exists yet — every iteration is a human action.

LEVEL 2

Operator
Hands-on. Agents run, but manually — a person kicks off each run, watches it, and intervenes. Most organizations evaluating agentic AI today sit here.

LEVEL 3

Loop Engineer
Proactive. Loops are designed deliberately — discovery through verification, with real quality gates and bounded autonomy. The agent runs unattended within defined limits.

LEVEL 4

System Architect
Strategic. Builds self-improving agent ecosystems — fleets of orchestrated agents that capture lessons from failures and feed them back into future runs automatically.

Most enterprises today sit at Level 1 or 2 — capable of running an agent, but not of designing a loop that runs unattended within real guardrails. The jump to Level 3 is what this guide is built to help an architect make: it's where Sections 04–09 stop being theory and start being a checklist.

SECTION 13, CONTINUED

The Self-Learning Loop

What Level 4 looks like in practice: a loop that gets measurably smarter every time it fails.

The defining trait of a Level 4 organization is not that its agents make fewer mistakes — it's that each mistake is made exactly once. The system that enforces this is itself a closed loop, running on top of every other loop the agent executes.



The mechanism that matters is step 4, not step 3. Plenty of teams capture a postmortem and never act on it. The discipline that separates Level 4 from Level 3 is making the lesson a durable, machine-readable input the agent actually reads on its next run — not a document a human has to remember to re-read and manually re-explain.

This is also where the building blocks from Section 06 compound: **memory** is what holds the lesson, **skills** are what make it reusable across tasks, and the **quality gate** from Section 07 is what catches the mistake in the first place. A self-learning loop is not a separate system — it's the same architecture in this guide, run continuously, with one extra step added at the end of every cycle.

SECTION 14

Implementation Roadmap & Architect's Checklist

A 90-day path from Level 1 to a single governed production loop, plus the checklist to run before any agent gets write access to a real system.

Days 1-30 — Foundation

- Pick one bounded, low-risk workflow
- Choose the simplest pattern that fits (Section 04)
- Define success criteria and a hard budget before writing any prompt
- Stand up structured logging from day one

Days 31-60 — Closing the loop

- Build the quality gate before expanding scope
- Add the maker/checker subagent split
- Move durable context into external memory
- Run adversarial tests against the tool surface

Days 61-90 — Governed production

- Assign the workflow an explicit autonomy tier (Section 09)
- Wire gateway-level access controls, not prompt-level instructions
- Document the system for your applicable regulatory framework
- Schedule recurring evaluation, not just launch-day testing

Pre-launch checklist

- ☐ The loop has an explicit, written success condition and a hard termination condition
- ☐ Token, cost, and iteration budgets are enforced by the orchestration layer, not requested of the model
- ☐ Every tool the agent can call has a documented, tested interface and least-privilege scope
- ☐ A quality gate exists that the agent cannot self-certify past
- ☐ The agent that does the work and the agent that checks the work are different instances
- ☐ Every action is logged with enough detail to reconstruct the decision chain after the fact
- ☐ The autonomy tier is assigned deliberately, matched to the reversibility of the action
- ☐ A human owns override authority, and the path to exercise it is tested, not theoretical
- ☐ The system has been adversarially tested for prompt injection and tool misuse
- ☐ Applicable regulatory obligations (Section 10) have been reviewed with legal counsel

Start simple, measure everything, add complexity only when it delivers measurable value. The best architecture is the simplest one that meets today's requirements — not the most sophisticated one you can build. Composability means today's single-purpose loop can become tomorrow's orchestrated fleet without a rewrite, as long as the foundational discipline in this checklist was there from the start.

APPENDIX

Sources & Further Reading

This guide synthesizes primary engineering sources, current governance research, and 2026 enterprise survey data. Figures and regulatory deadlines should be reverified before being used in a compliance filing.

- **Anthropic — "Building Effective Agents."** The foundational reference for the five composable workflow patterns and the workflows-vs-agents distinction used throughout Sections 04–05. anthropic.com/engineering/building-effective-agents
- **Anthropic — "How We Built Our Multi-Agent Research System."** Source for the orchestrator/lead-agent/subagent reference design discussed in Section 05. anthropic.com/engineering/multi-agent-research-system
- **NIST — AI Risk Management Framework & CAISI Agent Initiative (2026).** Source for the Govern/Map/Measure/Manage structure and the agent-specific profile discussed in Sections 10–11.
- **EU AI Act — High-Risk System Obligations.** Source for human oversight, traceability, and documentation requirements discussed in Section 10.
- **NextAgile — "Agentic AI Architecture Framework for Enterprises in 2026."** Source for adoption statistics and ERP/CRM/ITSM/HR integration patterns in Sections 01 and 09.
- **Elevate Consult — "Agentic AI Security & Governance in 2026" and "Agentic AI Security: How to Govern Autonomous Agents."** Source for the security risk taxonomy and governance recommendations in Section 11.
- **CertMage — "Agentic AI Governance Frameworks 2026."** Source for the human-in-the-loop vs. human-on-the-loop framing in Section 09 and Colorado AI Act deadline in Section 10.
- **Elementum AI — "Human-in-the-Loop Agentic AI" and "AI Governance Framework: Enterprise Guide for 2026."** Source for compliance-as-architecture framing referenced in Sections 09–10.
- **Covasant — "EU AI Act Compliance for Autonomous AI Agents in 2026."** Source for the "glass box" framing and minimum compliance checklist in Section 10.
- **Oracle Developers — "What Is the AI Agent Loop?"** Source for the tool-calling control flow described in Section 02.

This guide was produced by Claude (Anthropic) by synthesizing publicly available engineering documentation and 2026 industry research current as of the publication date. It is intended as an architectural and educational reference, not as legal, regulatory, or compliance advice. Verify all regulatory deadlines, statistics, and framework details against primary sources before relying on them for a compliance decision.